

200 Gb/s PolMux Link with 32 dB optical power budget, 28GHz electrical bandwidth and LO reuse for VHS-PONs

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Abstract *In this work we experimentally demonstrate a 25km long, 200Gb/s PolMux multiCAP downstream link for Very-High Speed PONs, using a narrow 28GHz electrical bandwidth and obtaining an optical power budget of 32dB while reusing the LO at the receiver as the upstream source.*
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Introduction

Optical access networks are increasingly considered as a promising option for the development of high throughput digital fronthauls in 5G and 6G networks [1]. Current standards of Passive Optical Networks (PONs) as High-Speed PONs (HSP) have been developed at 50 Gb/s line rate for downstream links [2], so next generation very-high speed PONs (VHSP) are likely to transmit downstream data to users at bit rates of up to 200 Gb/s. Optical standards for 200 Gb/s per wavelength are already being developed for their use in PONs, datacentres and in high speed short links [3]. In this context, coherent PON is being explored as a potential technology for VHSP, due to the increase in receiver sensitivity, Chromatic Dispersion (CD) management and superior spectral efficiency, enabled by a higher bit per baud [4]. However, the polarization sensitivity inherent to coherent reception, the increase in the cost of the Optical Network Units (ONUs) due to the addition of coherent receivers, and the higher energy consumption due to DSP are the main remaining issues for a broad adoption of this technology. However, relevant DSP is also expected for 50G and VHSP IM/DD based solutions [5].

In previous works, we have addressed these problems through developing polarization-independent, low-cost heterodyne receivers requiring only one coupler, one Polarization Beam Splitter (PBS) and two single Photo-Detectors (PDs) [6]. The higher bandwidth demand of heterodyne detection has been mitigated using efficient Optical Single-SideBand (OSSB) modulation, which, in addition, solves CD issues. Moreover, we demonstrated that the same architecture and components can support polarization-multiplexed (PolMux) transmission [7], doubling the bit rate at the cost of a modest increase in DSP for polarization demultiplexing.

In this work, we present a downstream scheme based on multiband Carrierless Amplitude and Phase (multiCAP) and OSSB

modulations, capable of achieving 200 Gb/s over 25 km using less than 30 GHz electrical bandwidth. Compared to conventional PAM4 (Pulse Amplitude Modulation) IM/DD systems, our approach reduces bandwidth usage by a factor of four, and by half compared to other 200 Gb/s coherent PON proposals [8], while requiring only one coupler, one PBS and two 30 GHz PDs. Finally, we have used the LO at the ONU to serve as the optical source for a 20 Gb/s upstream link to measure its impact in the downstream link and as a proof of concept for a future low-cost coherent ONU with just one laser emitting using IM.

Experimental setup

Fig. 1 shows the experimental setup used to evaluate the performance of the 200 Gb/s downstream link employing a PolMux coherent receiver for the 200 Gb/s downstream link, and a standard IM/DD for the 20 Gb/s upstream link.

In the downstream link, a Tuneable Laser Source (TLS) operating at 1550 nm is modulated using a dual polarization IQ modulator. The electrical modulating signal is generated using a 4-channel 128 GSa/s Arbitrary Waveform Generator (AWG), with one pair of outputs dedicated to each polarization. The signal consists of multiCAP spectral configuration to achieve an

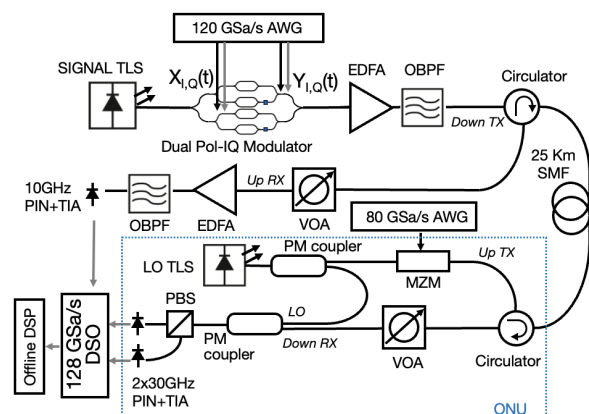


Fig. 1: Experimental setup.

aggregated data rate of 200 Gb/s. The configuration uses 5 multiCAP bands, all with identical settings: 5 Gbd and 16-QAM symbol mapping, yielding 20 Gb/s per band and polarization for an aggregated bit rate of 200 Gb/s. It is worth noting that this same bit rate can be obtained with a different set of bands and bitrate per band. The multiCAP signals are digitally Hilbert transformed, obtaining the I and Q components required for generating OSSB modulation with the IQ modulator, which is set at its minimum optical power value. The high-resolution optical spectrum for one polarization of the transmitted data, coupled with the LO at the receiver, is depicted in Fig. 2 (a). This spectrum clearly displays the multiCAP bands and the OSSB modulation for the downstream configuration as well as the carrier and the LO. The optical spectrum is 28 GHz for 5 bands, which have been power loaded to obtain a flatter optical power per band in the receiver.

The PolMux optical signal is amplified by an EDFA and filtered with a 100 GHz Optical Band Pass Filter (OBPF) to mitigate the EDFA ASE noise, before being injected into a 25 km spool of standard single-mode fibre (SSMF). The launch power of the signal at the input of the SSMF is +7 dBm, i.e., +4 dBm per polarization. A Variable Optical Attenuator (VOA) is included prior to the receiver to adjust the received power. There are no polarization controllers in the setup due to the polarization insensitiveness of the receiver.

The coherent receiver couples the received signal with the LO, which is provided by a TLS whose SOP is fixed at 45° relative to the PBS axes. The combined field is separated into two orthogonal polarizations that are detected by two 30 GHz PIN-TIA photodetectors. The LO is shifted 0.22 nm or 28 GHz with respect to the carrier, as can be seen from Fig. 2(a). This LO laser is splitted in two arms to be also used in the

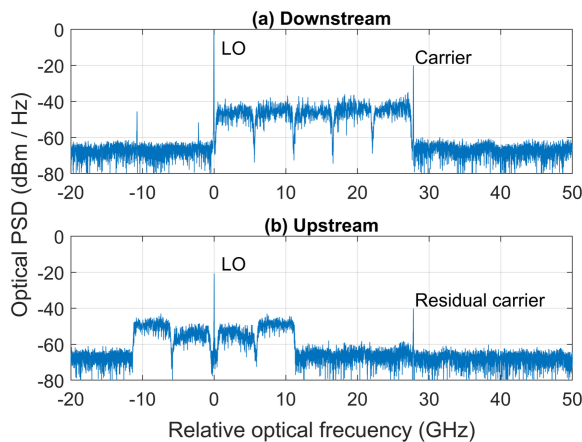


Fig. 2: High resolution optical spectra of the (a) downstream and (b) upstream channels transmitted through the 25 km of SSMF with the opposite-direction channel active.

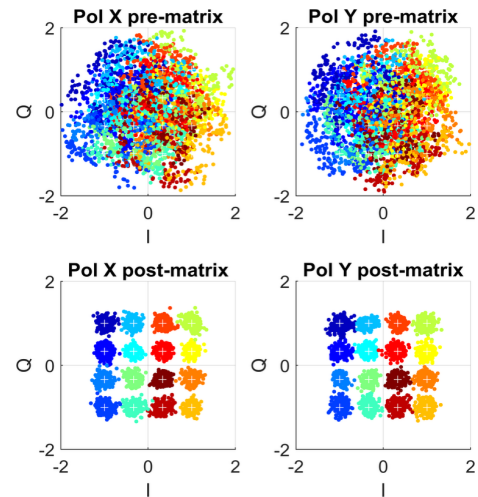


Fig. 3: PolMux signals at the receivers before and after the DSP polarization demultiplexing.

upstream channel. The LO optical power at the receiver is approximately 8 dBm. The electrical outputs of the photodetectors are digitized using a 40 GHz Digital Storage Oscilloscope (DSO) for offline DSP. This DSP includes synchronization by a PLL, multiCAP filtering, IQ sampling, polarization demultiplexing, equalization by a decision feedback equalizer (DFE), and symbol decoding to compute error vector magnitude (EVM) and bit error rate (BER). Fig. 3 shows the constellations received before and after applying the rotation matrix to obtain the two polarization multiplexed channels.

In the upstream channel, the optical source is the same LO laser, which is modulated in IM using a standard Mach Zehnder Modulator (MZM). Two 5 Gbd multiCAP QPSK DSB bands are transmitted for a bit rate of 20 Gb/s. The signal is attenuated by a VOA and received by an EDFA preamplified IM/DD photoreceiver. The signal is acquired by another channel of the same DSO and processed without any DSP. The high-resolution spectrum of the upstream channel can be appreciated in Fig. 2(b). The second multiCAP band must be power loaded to compensate the losses produced at higher frequencies due to chromatic dispersion.

Results

The sensitivity curves of the received 200 Gb/s downstream link, obtained without the upstream link active, are shown in Fig. 4. They were measured under three scenarios: a back-to-back (BTB) connection and a propagation over 25 km of SSMF with the VOA placed before the coherent receiver (as in Fig.1) or before the 25 km of SSMF. The required optical power of the received signals to achieve a BER below 2×10^{-2} , which corresponds to the reference threshold for soft decision low density parity check (SD-LDPC)

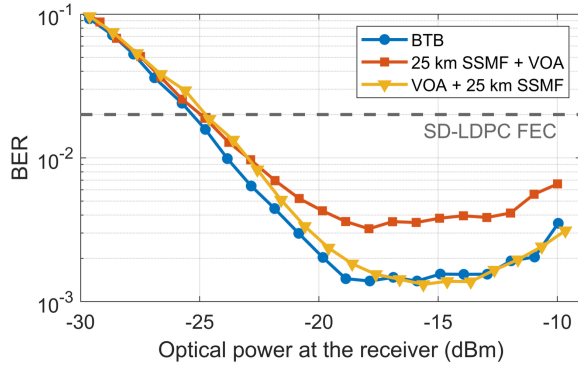


Fig. 4: Sensitivity curves for downstream 200 Gb/s link for BTB and 25km fibre propagation, with the VOA placed before or after the fibre.

FEC, is around -25 dBm for the three scenarios, giving rise to a 32 dB optical power budget. In the figure, it can be seen that BER floors appear at received optical powers of around -18 dBm. They are caused by the interference between the desired heterodyne detection with the direct-detected signal and because of the phase distortion of the MultiCAP signals due to the nonlinearities produced by the optical power injected into the 25 km SSMF. We can observe that if the VOA is placed before the pool of SSMF, i.e. the input optical power into the fibre is reduced, the floor decreases until the BTB case.

The propagation penalty introduced by 25 km of SSMF is less than 0.5 dB for both analysed cases. This low penalty confirms the mitigation of the chromatic dispersion by the OSSB modulation, avoiding the need of DSP-CD compensation.

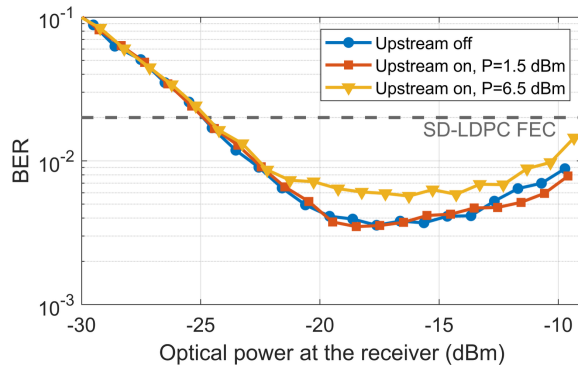


Fig. 5: Sensitivity curves for the downstream link with the upstream channel turned off and turned on with two different optical power values

Fig. 5 shows the sensitivity of the downstream channel when the upstream is transmitted with two different optical powers, +1.5 and +6.5 dBm. These two optical powers have been obtained by inserting a fixed attenuator at the input, as the change in the LO optical power impacts in the downstream coherent receiver. It can be seen that, for the lower input upstream optical power, the sensitivity at 2×10^{-2} BER value remains around -25 dBm. When the optical power

increases to +6.5 dBm, the floor increases due to the Rayleigh scattering of the upstream channel, although the sensitivity at the 2×10^{-2} BER value remains the same. In Fig. 2(a) a clear Brillouin peak produced by the LO in the upstream can be seen in the downstream spectrum, although it is out of the received band frequencies.

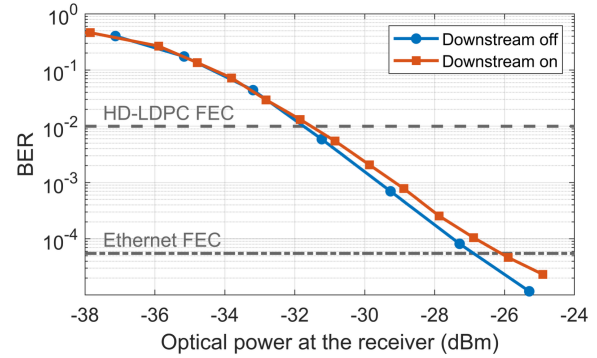


Fig. 6: Sensitivity curves for the 20Gb/s upstream channel for 25km SSMF propagation with the downstream channel turned on and off

Finally, Fig. 6 shows the sensitivity curves of the upstream channel after 25 km of SSMF transmission at +1.5 dBm, with the downstream channel either active or inactive. The +7 dBm optical power inserted into the downstream channel does not affect the sensitivity at higher BER values of 1×10^{-2} , corresponding to HD-LDPC FEC, which is around -32 dBm. This fits within the 32 dB of optical power budget of the downstream channel. If we look at the sensitivity of the upstream transmission at 5×10^{-5} BER, which corresponds to Ethernet KR1-FEC, there is a 1 dB penalty in the upstream transmission. Under -25 dBm the BER is below the 1×10^{-5} value, which is our BER limit of detection. In Fig. 2(b) a residual value of the carrier produced by the Rayleigh scattering of the downstream can be seen. Again, it does not seem to affect the upstream although two of the bands are sharing the bandwidth around the LO.

Conclusions

We have demonstrated experimentally a 200 Gb/s downstream link over 25 km of SSMF using a cost-effective PoIMux heterodyne coherent receiver and 20 Gbd multiCAP signals. Thanks to OSSB modulation, both optical and electrical bandwidths remain below 28 GHz. 32 dB of measured optical power budget has been confirmed. We have reused the LO of the ONU to act as the upstream optical source, which can transmit IM signals up to 6.5 dBm without affecting the minimum sensitivity of the channel, although the downstream signal presents higher BER floors due to Rayleigh scattering and phase distortion of the multiCAP signals.

Acknowledgements

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References

- [1] W. Hu and Y. Zhu, "100G and Beyond for PON and Short Reach Optical Networks," *Optical Fiber Communication Conference (OFC) 2023*, Paper Th3G.5, 2023. DOI: [10.1364/OFC.2023.Th3G.5](https://doi.org/10.1364/OFC.2023.Th3G.5)
- [2] "G.9804.3: 50-Gigabit-capable passive optical networks (50G-PON): PMD layer specification," ITU, 2021, <https://www.itu.int/rec/T-REC-G.9804.3-202109-I>
- [3] D. Zhang, D. Liu and D. Nessel. "Progress of Very High Speed PON in ITU-T", *Journal of Optical Communications and Networking*, vol. 18, no. 2, pp. A112-A122, 2026, DOI: [10.1364/JOCN.573747](https://doi.org/10.1364/JOCN.573747)
- [4] Z. Jia, H. Zhang, K. Choutagunta, and L. Campos, "Coherent passive optical network: applications, technologies, and specification development," *Journal of Optical Communications and Networking*, vol. 17, no. 1, pp. A71-A86, 2025. DOI: [10.1364/JOCN.535200](https://doi.org/10.1364/JOCN.535200).
- [5] D. Che and X. Chen, "Modulation Format and Digital Signal Processing for IM-DD Optics at Post-200G Era," *Journal of Lightwave Technology*, vol. 42, no. 2, pp. 588-605, 2024, DOI: [10.1109/JLT.2023.3311716](https://doi.org/10.1109/JLT.2023.3311716)
- [6] M. Barrio, D. Izquierdo, J. A. Altabás, and I. Garcés, "50 Gb/s Transmission using OSSB-MultiCAP Modulation and a Polarization Independent Coherent Receiver For Next-Generation Passive Optical Access Networks," *J. Journal of Lightwave Technology* vol. 39, pp. 5722–5729, 2021. DOI: [10.1109/JLT.2021.3092951](https://doi.org/10.1109/JLT.2021.3092951)
- [7] D. Izquierdo, M. Barrio, P. Sevillano, J. A. Altabas and I. Garcés, "100 Gb/s/λ polarization multiplexing PON downlink based on simplified heterodyne coherent reception," *Journal of Optical Communications and Networking*, vol. 16, no. 12, pp. 1241-1248, 2024, DOI: [10.1364/JOCN.530478](https://doi.org/10.1364/JOCN.530478)
- [8] I. B. Kovacs, M. S. Faruk, P. Torres-Ferrera and S. J. Savory, "Simplified coherent optical network units for very-high-speed passive optical networks," *Journal of Optical Communications and Networking*, vol. 16, no. 7, pp. C1-C10, 2024, DOI: [10.1364/JOCN.514867](https://doi.org/10.1364/JOCN.514867)